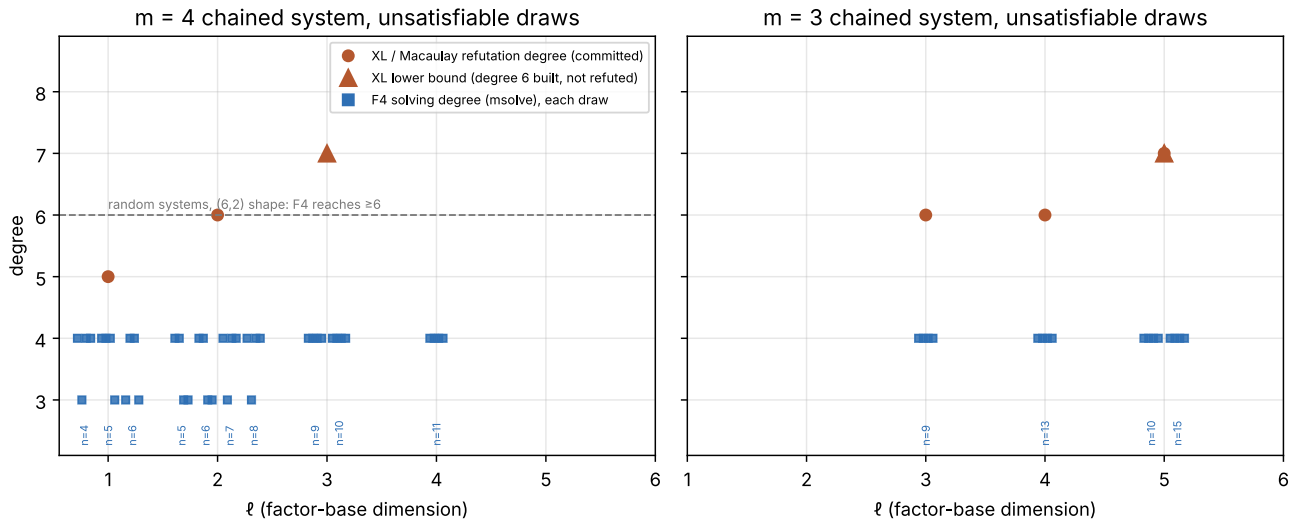
Item 6: the $m = 4$ Weil-descent degree cells, and equivariant F4

Campaign `research/ic_next10_20261014/`, item "### 6." of `reports/Index calculus leads next steps.md`. The protocol is frozen in [PROTOCOL.md](), written before any main-arm data. Class: **stage diagnostic**. Nothing here computes an S or a rho ratio, and every $n = 131$ figure is an extrapolation.

Verdict

POSITIVE (levels off) at $(4, 4)$ and at $(15, 5)$. BLOCKED at $(4, 5)$ and $(4, 6)$ by the 3 GB memory cap. The kill criterion is not met.

- **The F4 solving degree does not grow with ℓ.** msolve's F4 refutes every unsatisfiable draw it finished at a maximum step degree of **4**, and some at 3. That covers 56 draws:
  - $m = 4$: $\ell = 1, 2, 3, 4$, including all four $(n, \ell) = (11, 4)$ draws;
  - $m = 3$: $\ell = 3, 4, 5$, including the four committed $(15, 5)$ draws.
- **The committed rising degrees are XL degrees, not F4 degrees.** Every degree committed so far (5, 6, 6–7, 7, ≥7) is the refutation degree of bounded Macaulay (XL) linear algebra. F4 reuses its degree falls, and on the **same draws** it reads 4 where XL reads 5, 6, 7 and ≥7.
  - The kill criterion was written in those XL degrees: "≥8 at $(4, 4)$ and ≥8 at $(15, 5)$". On F4 it is far from met, at 4 and 4.
  - Its other half, "≤7 at both $(4, 4)$ and $(4, 5)$", holds at $(4, 4)$. $(4, 5)$ is BLOCKED, so the formal "reopen" condition is half-met.
- **The instrument can read degrees above 4.** On random cubic systems with the same shape as the $m = 4$ cells, the same instrument reaches degree 6 at the $(6, 2)$ shape and 5 at the $(9, 3)$ shape. Those runs were cut off at 300 s; the real systems finish at degree ≤4 in 0.1 s.
- **Where the wall is.** At $(13, 4)$ (42 unknowns) and $(13, 5)$ (46 unknowns), msolve exceeds the 3 GB address-space cap and segfaults on a failed hash-table enlargement. This happened on six attempts, at 335–839 s and peak RSS 1.6–1.8 GB, while still in degree-4 rounds. These runs are a resource limit, not evidence either way.
  - A satisfiable-draw substitute at $\ell = 5$, $(10, 5)$ with 40 unknowns, hit the same wall.
  - $\ell = 6$ was not attempted. Its smallest unsatisfiable-capable n gives $N \geq 54$, past the cap.
- **G2 equivariant F4: finished as one complete Macaulay pass.** The pass takes block RREF over F_{2^d}, applies the inverse transform back to F_2 rows, and recovers the degree-fall rows. Its reduced row echelon form is **bit-identical** to the full F_2 RREF at $n = 7, 11, 13, 17$.
  - The equivariant pass and the full F_2 pass are timed in the same process; §3 gives the times: 44×, 58× and 73× in CPU at n = 11, 13, 17.
  - It is not wired into `koblitz_groebner::reduce_system`, and it is not an iterated GB computation. See "What did not get done".



F4 against XL, by ℓ

1. The degree cells

msolve 0.6.5, `-t 1 -g 2 -v 2`, DRL, characteristic 2, with the field equations `x_i² + x_i` appended. `D_F4` is the maximum `deg` over all F4 rounds of the run (`run_f4.py` parses the verbose log). The systems are exported by `examples/ic_next10_06_chain_export.rs`, which replays `dreg_ladder`'s cell seed, sampling and exact solution count. 48 of the 52 calibration draws are the **committed** draws, matched on `v_basis` and `x_r` (`runs/c2_identity.txt`). The other four, `m = 3 (9, 3)` at seed 20260925, are new draws of a committed cell.

Main cells (seed 20261014) and calibration on committed draws:

m	(n, ℓ)	N	S	XL refutation degree (committed)	F4 degree per draw	GB = [1]	wall s	peak RSS MB	largest F4 matrix
4	(4, 1)	12	+0	5 5 5 5	4; 3; 4; 4	yes	0.1–0.1	12–12	171 × 210
4	(5, 1)	14	+1	5 5 5 5	4; 4; 4; 3	yes	0.1–0.1	12–12	549 × 590
4	(6, 1)	16	+2	5 5 5 5	3; 4; 4; 3	yes	0.1–0.1	12–12	362 × 440
4	(5, 2)	18	-3	6 6 6 6	4; 4; 3; 3	yes	0.1–0.2	12–12	2,617 × 2,495
4	(6, 2)	20	-2	6 6 6 6	4; 4; 3; 3	yes	0.1–0.1	12–12	2,753 × 2,558
4	(7, 2)	22	-1	6 6 6 6	4; 3; 4; 4	yes	0.1–0.3	12–12	4,620 × 4,321
4	(8, 2)	24	+0	6 6 6 6	4; 3; 4; 4	yes	0.1–0.6	12–22	7,836 × 8,043
4	(9, 3)	30	-3	≥7 ≥7 ≥7 ≥7	4; 4; 4; 4	yes	2.0–4.2	57–83	16,684 × 15,037
4	(10, 3)	32	-2	≥7 ≥7 ≥7 ≥7	4; 4; 4; 4	yes	9.3–43.4	116–249	36,249 × 26,375
4	(11, 4)	38	-5	not measured	4; 4; 4; 4	yes	386.3–804.8	1047–1478	84,355 × 66,607
4	(13, 4)	42	-3	not measured	u0: cap, deg ≤4 so far; u1: cap, deg ≤4 so	NO	369.8–593.0	1689–1809	92,364 × 102,107

m	(n, ℓ)	N	S	XL refutation degree (committed)	F4 degree per draw	GB = [1]	wall s	peak RSS MB	largest F4 matrix
					far; u0 -u 1: cap, deg ≤4 so far				
4	(13, 5)	46	-7	not measured	u0: cap, deg ≤4 so far; u1: cap, deg ≤4 so far; u0 -u 1: cap, deg ≤4 so far	NO	335.5–838.5	1628–1808	105,669 × 149,138
3	(9, 3)	18	+0	6 6 6 6 (other draws)	4; 4; 4; 4	yes	0.1–0.2	12–12	1,551 × 1,740
3	(13, 4)	25	+1	6 6 6 6	4; 4; 4; 4	yes	3.5–4.3	40–43	15,213 × 15,481
3	(10, 5)	25	-5	7 7 7 7	4; 4; 4; 4	yes	6.9–23.2	70–97	21,978 × 15,430
3	(15, 5)	30	+0	≥7 ≥7 ≥7 ≥7	4; 4; 4; 4	yes	56.7–77.3	247–322	38,902 × 31,658

The `(13, ·)` rows are attempts, not measurements: every one crashed at the 3 GB cap (exit −11) while still in degree-4 rounds.

Secondary no-count arm:

- m=4 (n=10, ℓ=5) N=40 draw 0: 3 GB cap hit, GB [1]? False, GB length None, max F4 degree 4, 911 s wall, 1725 MB, largest matrix 67,622 × 77,361
- **Unchanged in `n` and in `ℓ`.** `D_F4` ∈ {3, 4} throughout, with every cell's median at 4 or 3.5.
- **The XL degree rises over the same draws:** `m = 4` goes 5, 6, ≥7, and `m = 3` goes 6, 6, 7.
- **What does grow is the matrix and the memory.** The largest F4 matrix goes from 2.7k × 2.6k at `(6, 2)` to 84k × 67k at `(11, 4)`. That is about `C(38, ≤4)` = 82,993 columns, the full degree-4 monomial count.
    - Peak memory goes 12 MB, then 83 MB at `(9, 3)`, then 1.0–1.5 GB at `(11, 4)`.
    - The cost is set by the size of the degree-4 Macaulay space, not by a rising degree.

Why XL and F4 differ (mechanism, standard)

- The XL refutation degree `D_XL` asks whether `1` lies in the span of `{t · f_i : deg(t · f_i) ≤ D}`, the original equations times monomials.
- F4 at maximum degree `D` multiplies **every** polynomial it has found by monomials, including the degree falls. The committed first fall degree is 3 on every draw. F4 reduces those falls and multiplies them by degree-1 and degree-2 monomials, so it stays at degree 4.
- XL instead has to reach the same products through the original equations, at degree `3 + (D − 3)` plus the multiplier depth.
- So `D_F4 ≤ D_XL` always. On this system the gap is 1–3 degrees, and it widens with `ℓ`: it is 1–2 at `ℓ = 1`, 2–3 at `ℓ = 2`, and ≥3 at `ℓ = 3`.
- H2 of the protocol holds on every one of the 48 paired draws.

Controls

- **C1, the XL positive control.** The repo's frozen instrument, `dreg_ladder` (sha256 in `runs/dreg_ladder.sha256`), was rerun on committed draws:
    - `m = 4 (6, 2)` u0–u3 refutes at **6 6 6 6**, as committed;
    - `(9, 3)` u0 reads **≥7** (degree 6 built in full, 1,409 s, 342 MB), as committed;
    - files: `runs/c1_xl_*.jsonl`.
- **D, sensitivity.** The random systems keep each template equation's term counts per degree, with uniformly random monomials (`gen_random.py`, seeds 1–4). Semi-regular `D_reg` is 7 at the `(6, 2)` shape and 8 at the `(9, 3)` shape.
- r62-1: max F4 degree reached 6 (timeout at 300 s), largest matrix 134,797 × 138,588, peak 614 MB
- r62-2: max F4 degree reached 6 (timeout at 300 s), largest matrix 143,307 × 144,597, peak 728 MB
- r62-3: max F4 degree reached 6 (timeout at 300 s), largest matrix 134,378 × 137,198, peak 724 MB
- r62-4: max F4 degree reached 6 (timeout at 300 s), largest matrix 140,453 × 144,040, peak 645 MB
- r93-1: max F4 degree reached 5 (timeout at 300 s), largest matrix 161,774 × 286,849, peak 646 MB
- r93-2: max F4 degree reached 5 (timeout at 300 s), largest matrix 159,673 × 284,683, peak 678 MB
- **E, instrument correctness.** On every unsatisfiable draw the reduced GB is `[1]`. On every satisfiable draw it is not:
    - m=4 (9,3) draw 3: exact count 12, GB [1]? False, GB length 47, F4 degree 4, 11 s
    - m=4 (10,3) draw 0: exact count 12, GB [1]? False, GB length 49, F4 degree 4, 158 s
    - m=4 (10,3) draw 3: exact count 4, GB [1]? False, GB length 35, F4 degree 4, 114 s
    - m=4 (11,4) draw 0: exact count 154, GB [1]? False, GB length 594, F4 degree 5, 496 s
    - m=4 (11,4) draw 1: exact count 12, GB [1]? False, GB length 56, F4 degree 4, 428 s
    - Pass.
    - A satisfiable `(11, 4)` draw with 154 solutions needs degree **5** to finish its full GB. That is a solving degree, not a refutation degree. It shows again that the instrument reports more than 4 when the computation needs more.

2. What it means for the 106-bit spread (derived, not measured)

`cost_d4.py` reruns `examples/descent_degree_cost`'s model (same formulas; its `D = 6` rows reproduce the committed `2^76.31` / `2^+15.50` and `2^88.05` exactly) at the degrees F4 shows. The output is `cost_d4-output.txt`.

m	D	w	best ℓ	total	vs rho (2^60.81)
4	4	1	33	2^68.43	2^+7.62
4	4	2	34	2^93.91	2^+33.10
4	3	1	31	2^66.15	2^+5.34
4	6	1	34	2^76.31	2^+15.50 (committed row)
3	4	1	40	2^82.27	2^+21.46

- **Under a constant F4 degree of 4, the `m = 4` route sits about 7.6 bits above rho at `w = 1`.** That is in unconverted column touches against group operations, so $\log_2 K$ comes off that gap.
    - That is 8 bits below the "levelled at 7" row the report quotes (`2^+21.30`), and about 165 bits below the $\ell + 4$ growth row (`2^+173.13`).
    - On this evidence the growth model is withdrawn for F4, and the spread narrows to "constant degree 4 (measured to $n \le 15$, $\ell \le 5$)".

- **This rests on extrapolation.** It assumes the F4 degree stays at 4 from $\ell \le 4$ ($m = 4$) and $\ell \le 5$ ($m = 3$) out to $\ell \approx 33$ and $n = 131$.
  - It also assumes F4's matrices stay near the full degree-4 column count, which they did at `(11, 4)`.
  - It is not measured.
  - At $w = 1$ the gap is still positive. So even before K, this does **not** show the route beating rho.
- **K (column touches per GF(2^131) group operation) was not measured.** See "What did not get done".

3. G2: equivariant F4 as a full Macaulay pass

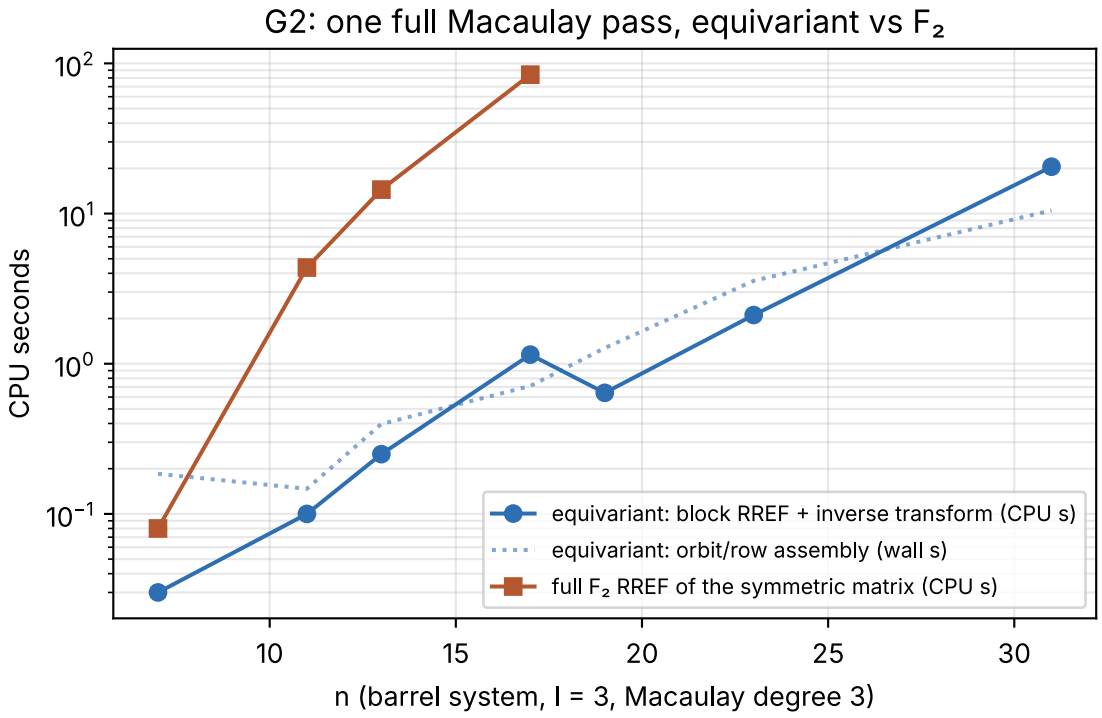
`examples/ic_next10_06_g2_full_pass.rs` copies `g2_wide_probe`'s barrel system and block construction and adds four things.

1. **Degree-descending column order inside every block.**
2. **Full RREF of each block:** the trivial block over F_2 and the character blocks over F_{2^d}.
3. **The inverse transform.** A reduced trivial-block row w maps to $u[\sigma^t c] = w[c]$. A character-block row maps, for each λ in an F_2-basis of F_{2^d}, to $u_\lambda[\sigma^t c] = \mathrm{Tr}(\lambda \cdot w[c] \cdot \zeta^{jt})$.
4. **The agreement checks.** Under `--control`:
   - the full F_2 RREF of the symmetric matrix is computed under the same column order;
   - the RREF of the transformed rows is computed;
   - the two must be bit-identical, which is equivalent to row-space equality;
   - the degree-fall row count and the refutation bit must also agree.

n	d	symmetric matrix	row orbits	block ranks (χ×orbit: rank)	rank	degree-fall rows	assembly s (wall)	blocks RREF + inverse (CPU s)	full F_2 RREF (CPU s)	ratio (CPU, without / with assembly)	RREF identical
7	3	5,827 × 10,382	889	χ1×3:624, χ3×3:621	4,406	489	0.18	0.03	0.08	3× / 0×	**yes**
11	10	17,281 × 34,606	1,631	χ1×10:1232	13,595	670	0.15	0.10	4.37	44× / 18×	**yes**
13	12	26,404 × 54,752	2,092	χ1×12:1612	20,999	879	0.40	0.25	14.44	58× / 22×	**yes**
17	8	53,242 × 115,792	3,194	χ1×8:2528, χ3×8:2528	43,019	1,381	0.71	1.15	84.09	73× / 45×	**yes**
19	18	71,677 × 158,558	3,835	χ1×18:3064	58,259	1,674	1.27	0.64	not run (matrix 1.3 GB > cap)	—	—
23	11	120,379 × 273,262	5,297	χ1×11:4292, χ5×11:4292	98,759	2,344	3.57	2.11	not run (matrix 3.8 GB > cap)	—	—
31	5	275,191 × 645,566	8,941	χ1×5:7325, χ3×5:7329, χ5×5:7329, χ7×5:7328, χ11×5:7325, χ15×5:7329	227,200	8,995	10.50	20.51	not run (matrix 20.7 GB > cap)	—	—

- **Agreement.** At $n = 7, 11, 13, 17$ the transformed rows' RREF is bit-identical to the full F_2 RREF of the symmetric Macaulay matrix, under the same degree-descending order. The degree-fall row counts agree too (489, 670, 879, 1,381). Neither side refutes: the barrel system at degree 3 does not contain 1.
- **Rank identity at the new sizes.** At $n = 19, 23$ the ranks equal `g2_wide_probe`'s committed full-F_2 ranks: 58,259 and 98,759.

- **Speed: 44×, 58× and 73× at `n = 11, 13, 17`.** That is CPU time of block RREF plus inverse transform against the full F_2 RREF, in the same process.
  - It is 25–45× once the equivariant side's orbit and row assembly is charged.
  - The full side includes building its own rows.
  - The ratio grows with `n`, as `RESEARCH_G2_EQUIVARIANT_F4_SCOPE.md` predicted.
- **`n = 31` (`d = 5`, six character blocks).** The equivariant pass finishes in 10.5 s of assembly plus 20.5 CPU s on a 275,191 × 645,566 symmetric matrix.
  - The full F_2 control would need about 21 GB dense, so it is not run.
  - So `n = 31` has a measured equivariant time and no measured ratio.
  - It runs at `l = 3`, degree 3, not the scope note's "dimension 16".
- **What is transformed back differs by mode.** With `--control` (`n ≤ 17`) every reduced row is transformed back. Without it (`n ≥ 19`) only the degree-fall rows are, which is what a solver consumes. That explains the dip at `n = 19` in the figure.
- **Both sides are naive Gauss–Jordan.** The control is a plain bit-packed kernel, not M4RI. The blocks use log/antilog tables. This is a same-process comparison, not a tuned-engine claim.
- **Scope.** This is G2 as **one complete Macaulay/XL linear-algebra pass** on the Frobenius-symmetric barrel system: row space, degree-fall space and refutation bit, all recovered blockwise.
  - It is not an iterated F4 with pair selection.
  - It is not wired into `reduce_system`.
  - It does not apply to the random-subspace cells of §1, which have no σ symmetry. Those cells were run on msolve without G2.



G2 full pass

Deviations from the protocol

- **Four draws at `(4, 4)` only.** `(13, 4)` and `(13, 5)` crashed at the 3 GB cap on every attempt (two draws each, plus one rerun each with msolve `-u 1`). `ℓ = 6` was not attempted. The protocol's BLOCKED rule applies.
- **The FGHR arm was not run**, as the protocol stated in advance.
- **The secondary no-count arm** (`--no-count`, satisfiable draws allowed) was added after the `ℓ = 5` crashes. Its single run also hit the cap. The other three jobs were stopped to free CPU.
- **Wall times come from a machine at load 15–20.** Ten experiments shared four cores. Each wall time is a practicality note, and `cpu_secs` is recorded where the runner had it.

- **Pilots.** Three msolve runs preceded `PROTOCOL.md`, as disclosed there. Their draws are included in the arms, not counted twice. The first arm-D run (`r62-1`, about 6 min, killed by hand) is kept as `runs/d_pilot_r62-1_killed.log`. The recorded D runs used a fixed 300 s cap.
- **The first `e_sat` run used an exporter built before `--sat-ok` reported counts.** It is kept as `runs/work/e_sat_oldbinary_nocount.jsonl` and superseded by `runs/e_sat.jsonl`.

What did not get done

- **`(4, 5)`, `(4, 6)`, and `(13, 4)` at `m = 4`.** They need more than 3 GB for msolve. Rough sizing from `(11, 4)`'s 1.5 GB and its matrix growth suggests 4–8 GB per draw. That is a modest host, not the 128 GB-class host the XL degree-7 runs needed.
- **G2 as a `reduce_system` engine and as an iterated F4.** F4's leading-term bookkeeping (a monomial order) is not σ-equivariant. The pass here produces the row space, the degree-fall space and the refutation bit blockwise, and these are what XL and Macaulay steps consume. It does not produce an F4 pair-selection loop. No `src/` file was touched.
- **`K`, the column touches per GF(2^131) group operation.** Not measured.
- **The `n = 31`, dimension-16 timing against `macaulay_profile`.** Replaced by the block-only `n = 31` run above, at `l = 3`. See G2.

Reproduce

```
        cd /home/user/crypto
        cargo build --release --example ic_next10_06_chain_export --example ic_next10_06_g2_full_pass --
example dreg_ladder
        D=research/ic_next10_20261014/06-equivariant-f4-m4
        # F4 degree, one job per draw: m:n:ell:seed:unsat_index
        python3 -I $D/run_f4.py $D/runs/c2_calibration.jsonl 4:6:2:20261006:0 3:15:5:20260928:0   # ...
        python3 -I $D/run_f4.py $D/runs/m_main.jsonl 4:11:4:20261014:0 4:11:4:20261014:1 4:11:4:20261014:2
4:11:4:20261014:3
        python3 -I $D/run_f4.py $D/runs/e_sat.jsonl 4:9:3:20261006:3:sat 4:11:4:20261014:0:sat
        python3 -I $D/gen_random.py $D/runs/rand/template-m4-6-2-u0.ms 1 /tmp/r.ms && F4_TIMEOUT=300 python3
-I $D/run_f4.py out.jsonl rand:/tmp/r.ms:r62-1
        # XL positive control
        F4_F2_MAX_ROWS=50000000 F4_F2_MAX_COLS=50000000 KIC_SPARSE_DENSE_FINISH=1 KIC_SPARSE_F5=1
KIC_SPARSE_DENSE_BUDGET_MB=2500 \
          target/release/examples/dreg_ladder --m 4 --cells 6:2:8 --ffd-max 5 --seed 20261006 --unsat-index 0
        # G2 full pass
        target/release/examples/ic_next10_06_g2_full_pass --n 7,11,13,17 --l 3 --degree 3 --control
        target/release/examples/ic_next10_06_g2_full_pass --n 19,23,31 --l 3 --degree 3
        python3 -I $D/cost_d4.py; python3 -I $D/summarize.py; python3 -I $D/plot.py
```

Every job ran single-threaded under `nice` and `prlimit --as=3000000000`. The binaries' sha256 values are in `runs/binaries.sha256` and `runs/dreg_ladder.sha256`.